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CSE.

Ch-1A

Functions

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Def^n of f^n

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Domain and Range of a f^n

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Graph of different type of f^ns

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Continuity and differentiability

Double system Data chake verification

Recommended Books:

1. P.K. Bhattacharjee → Differential Calcus.
2. Abu Yusuf → 〃
3. Das & Mukherjee → 〃
4. Frank Ayres (schaum's outline series) → Calculas
5. H.S.C text books.
6. S.S.C → Higher maths.
7. Google & Khan Academy.

Functions

*** Definition of function:**

A function is a rule that associates, with each value of a variable x in a certain set, exac[tly] one value of another variable y. The variable y is called the dependent variable, and x is called the indep[en]dent variable.

Symbolically, we write, $y = f(x)$ or $y = \phi(x)$.

For example, $y = x^2$ or $y = 3x + 5$ or $A = \pi r^2$.

Note:- Function as a relation: Let X and Y be two non empty sets. Then a relation from X to Y with domain is called a function, if no two different ordered pa[irs] in f have the same first co-ordinates.

(1) Example:- Let, $x = \{0, 1, 4, 9\}$ and $y = \{0, 1, 2, -2, 3\}$

let, $f = \{(0,0), (1,1), (4,2), (4,-2), (9,3)\}$

Clearly two different ordered pairs namely $(4,2)$ and $(4,-2)$ have the same first co-ordinates and therefore f is a relation but not a function.

(ii) Let, $x = \{1, -1, 2, -2\}$ and $Y = \{1, 4, 5\}$.

and let, $g = \{(1,1), (-1,1), (2,4), (-2,5)\}$ then two no different ordered pairs have the same first co-ordin[ates] and therefore g is a function.

Domain and Range of a function:

Domain:- The domain is the set of all possible x-values (inputs)

Range: The range is the set of all possible y-values (outputs).

For example, consider the equation, $y = x^2$.
in the equation we assume that any real value of x is an allowable input. Since there is no restriction on x. Thus the equation defines a function $f(x) = x^2$.
with domain $-\infty < x < \infty$.

As x varies over the domain of the function $f(x) = x^2$; the values of $y = x^2$ vary over the interval $0 \le y < \infty$. So, this is the range of f.

Definition of different type of functions:

1. Single-valued function:
A function $y = f(x)$ is called a single-valued in a domain R in only one value of y corresponds to each value of x in R.

Example, $y = 2x+5$, $y = x^2$, $y = e^{x/2}$, $y = \ln x + 5$ etc.

2. Many valued function:- A function $y = f(x)$ is called a multivalued in a domain R if more than one value of y corresponds to each value of x in R.

For example, $y^2 = 4ax$, $y = \sin^{-1}x$, $y = \cos^{-1}x$.

Suppose, $y = \sin^{-1}x$; $x = \frac{1}{2}$; $y = 30^\circ, 150^\circ, 390^\circ, \ldots$ etc

3. **Explicit functions:** A function which is directly expressed in terms of the independent variable is called an explicit function. i.e, $y = \cos x$, $y = x^2 - 5x$ or $x = y^2 - 2y + 1$.

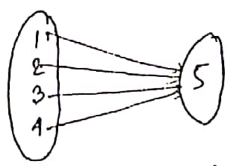
4. **Implicit function:** A function which is not expressed directly in terms of the independent variable is called an Implicit function.

$$x^3 + y^3 + 3axy = 0$$

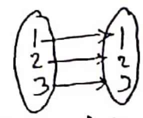
or, $F(x, y) = x^3 + 3x^2y + xy^2 + 5$ [ind. & dep. mix].

5. **Constant function:-** All elements of domain must be related with only one element of co-domain.

$f(x) = 5$ (constant)



6. **Identity function:-** If $f(x) = x$ then the function is called identity function.



7. **Periodic function:-** if $f(x) = f(x+T)$ for all values of x then f is called the periodic function. and T is called period. Trigonometrical functions such as $\sin x$ and $\cos x$ are periodic with 2π as period while

Explicit : A f^n which is directly expressed in term of the independent variable is called an implicit f^n.

e.g. $y = x\sin x$, $y = a\cos\theta$, $e^{ax}\cos bx$ etc.

Implicit : $x^2 + y^2 = a^2$,

$ax^2 + 2hxy + by^2 = 0$

$3x^2 + 2xy^2 + 4xy + 5x + 4y + 3 = 0$

$\tan x$ is periodic with period π.

$$\sin(2\pi + x) = \sin x \quad \& \quad \cos(2\pi + x) = \cos x.$$

8. <u>Odd or even function</u> A function $f(x)$ is said to be odd if $f(-x) = -f(x)$; and even if

$$f(-x) = f(x)$$

Examples: 1. $f(x) = x^2$; $f(-x) = (-x)^2 = x^2 = f(x)$ = even f^n

2. $f(x) = x^3$; $f(-x) = (-x)^3 = -x^3 = -f(x)$ = odd f^n

<u>Tips:-</u> Trigonometric f^ns: $y = \cos x$ & $y = \sec x$ are even functions, since, $y = f(x) = \cos x$

$\Rightarrow y = f(-x) = \cos(-x) = \cos x = f(x)$ = even.

and
$$\left.\begin{array}{l} y = \sin x \\ \;\;= \operatorname{cosec} x \\ \;\;= \tan x \\ \;\;= \cot x \end{array}\right\} \text{odd } f^n; \text{ since } y = \sin(-x) = -\sin x.$$

Test whether the following functions are odd or even:

(i) $f(x) = 1 - \frac{1}{x^2}$

(ii) $g(x) = x + \frac{1}{x}$.

(iii) $f(x) = \ln(x + \sqrt{1+x^2})$.

Solⁿ:-

(i) Given, $f(x) = 1 - \frac{1}{x^2}$ —— ①

putting $x = -x$ in eqⁿ(1) we get;

$f(-x) = 1 - \frac{1}{(-x)^2} = 1 - \frac{1}{x^2} = f(x)$ = even fⁿ

(ii) Given $g(x) = x + \frac{1}{x}$ —— ①

putting $x = -x$ in ①

$g(-x) = (-x) + \frac{1}{(-x)} = -x - \frac{1}{x} = -(x+\frac{1}{x}) = -g(x)$

Since $g(-x) = -g(x)$ therefore the given fⁿ is odd fⁿ.

(iii) $f(x) = \ln(x + \sqrt{1+x^2})$ —— ①

putting $x = -x$; in(1) we get

$f(-x) = \log(-x + \sqrt{1+(-x)^2})$

$= \ln(-x + \sqrt{1+x^2})$

$= \ln\left\{\frac{(-x+\sqrt{1+x^2})(x+\sqrt{1+x^2})}{x+\sqrt{1+x^2}}\right\}$

$= \ln\left\{\frac{(\sqrt{1+x^2})^2 - x^2}{x+\sqrt{1+x^2}}\right\}$

$= \ln\left(\frac{1}{x+\sqrt{1+x^2}}\right)$

$= -\ln(x+\sqrt{1+x^2}) = -f(x)$ = odd fⁿ.

One-one function:-

A function whose denoted by $f: A \to B$ where

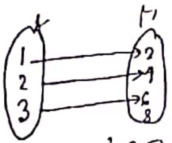
(I) Every domain has individual image in co-domain.

(II) But co-domain may be empty.

is called one-one function.

Example:- The function $f(x) = 2x$

then, $f: \{1, 2, 3\} \to \{2, 4, 6, 8\}$



4 [illegible]

Note: Co-domain:- A set containing the outputs called its co-domain.

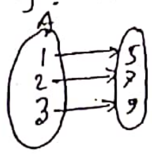
Onto function:

When a function i.e, (I) Every co-domain must be related with domain.

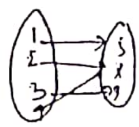
(II) But more than one domain can ~~not~~ be related with same co-domain.

Suppose, $y = f(x) = 2x + 3$.

then; $f: A = \{1, 2, 3\} \to B = \{5, 7, 9\}$



or.



[illegible]

Inverse function: If domain and co-domain are replaced to each other then it is called inverse function.

~~Suppose~~, $A = \{(-3,5), (-2,5), (-1,5), (0,5), (1,5), (2,5)\}$

$A^{-1} = \{(5,-3), (5,-2), (5,-1), (5,0), (5,1), (5,2)\}$

How do we find domains and Ranges.

Domain:

(I) If $f(x) = \frac{a}{0}$, $a \neq 0$, or $\frac{0}{0}$ or $a + ib$ $(b \neq 0)$

i.e, The denominator (bottom) of a fraction can not be zero.

(II) The number under a square root sign must be positive in this work. $f(x) = \sqrt{\phi(x)}$, $\phi(x) \geq 0$

(III) If $f(x) = \frac{\sqrt{\phi(x)}}{\sqrt{\psi(x)}}$ in this case

domain $= \phi(x) \geq 0$ and $\psi(x) \geq 0$.

Range:- The range is resulting y-values.

Find Domain and Range of the following functions:

(i) $f(x) = y = x+3$ —— (i)

equation (i) satisfies all the real values of x.

So, Domain $f = D_f = \{x : x \in \mathbb{R}\} = \mathbb{R}$

From (i) $x = y - 3$ —— (ii)

Again eq^n (ii) satisfies all the real values of x.

So, Range, $R_f = \{y : y \in \mathbb{R}\} = \mathbb{R}$.

$f(x) = 3x^2 + x + 2$ —— (i)

Sol^n:- Let $y = f(x) = 3x^2 + x + 2$ —— (ii)

eq^n (ii) satisfies all the real values of x, so

Domain, $D_f = \{x : x \in \mathbb{R}\} = \mathbb{R}$.

From (ii)

$$y = 3x^2 + x + 2$$

$$\Rightarrow 3x^2 + x + (2-y) = 0$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 3(2-y)}}{2 \cdot 3}$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{1 - 24 + 12y}}{6}$$

$$= \frac{-1 \pm \sqrt{12y - 23}}{6} \quad \text{—— (iii)}$$

equation (iii) satisfied,

$$12y - 23 \geq 0$$

i.e., $12y \geq 23$

$\Rightarrow y \geq \frac{23}{12}$

So, Range, $R_f = \{y : y \geq \frac{23}{12}\} = [\frac{23}{12}, \infty)$

$y = f(x) = \sqrt{x+3}$ —(1)

eq^n (1) satisfied if and only if $x+3 \geq 0$

i.e. $x \geq -3$

So, Domain, $D_f = \{x : x \geq -3\} = [-3, \infty)$

Range, $R_f = \{y : y \geq 0\} = [0, \infty)$

$y = f(x) = \sqrt{16 - x^2}$ —(1)

eq^n (1) satisfies when $16 - x^2 \geq 0$

$\Rightarrow x^2 \leq 16$

$\Rightarrow x \leq \pm 4$

$\Rightarrow -4 \leq x \leq 4$

Domain, $D_f = \{x : -4 \leq x \leq 4\} = [-4, 4]$

Range, $R_f = [0, 4]$.

$y = \sqrt{4 - x^2}$ —(1)

eq^n (1) satisfies when $4 - x^2 \geq 0$

$\Rightarrow x^2 \leq 4$

$\Rightarrow x \leq \pm 2$

$\Rightarrow -2 \leq x \leq 2$

Domain, $D_f = \{x : -2 \leq x \leq 2\} = [-2, 2]$

Range, $R_f = [0, 2]$

$y = \sqrt{x^2 - 16}$ — (1)

Here, $x^2 - 16 \geq 0$ or, $x^2 \geq 16$. So, Domain consists of the intervals $x \leq -4$ and $x \geq 4$.

$f(x) = \sqrt{\frac{1-x}{x}}$ —(1)

eqn (1) satisfies only when $1-x \geq 0$ and $x>0$

$\therefore$ Domain, $D_f = \{x : x>0 \text{ and } x \leq 1\} = (0, 1]$.

from (1) $y = \sqrt{\frac{1}{x} - 1}$

$x \in (0,1)$ & $y \in [0, \infty)$.

$y = f(x) = \frac{x-3}{2x+1}$ —(1)

Domain, $D_f = \mathbb{R} - \{-\frac{1}{2}\}$

Again from (1)

$$y = \frac{x-3}{2x+1}$$

$$\Rightarrow 2xy + y = x-3$$

$$\Rightarrow 2xy - x = -y-3$$

$$\Rightarrow x = \frac{-y-3}{2y-1}$$

Range, $R_f = \mathbb{R} - \{\frac{1}{2}\}$

$y = f(x) = x^2+1$.

$D_f = \mathbb{R}$

$R_f = [1, \infty)$.

$x>0 \Rightarrow f(x) = x^2+1$

$x<0 \Rightarrow f(x) = x^2+1$

$x=0 \Rightarrow f(x) = 1$

H.W:- $y = f(x) = \frac{x-1}{2x-3}$ [From H.S.C text books]

1. $f(x) = \frac{x}{|x|}$

Soln:- Given, $f(x) = \frac{x}{|x|}$

If $x = 0$, $f(x) = \frac{0}{|0|} = \frac{0}{0}$ which is undefined

Domain of this function, $D_f = \mathbb{R} - \{0\}$

Range: $f(x) = \begin{cases} \frac{x}{x} & \text{if } x > 0 \\ \frac{x}{-x} & \text{if } x < 0 \end{cases}$

$= \begin{cases} 1 & \text{when } x > 0 \\ -1 & \text{" } x < 0 \end{cases}$

Range of this f^n, $R_f = \{1, -1\}$.

\# $f(x) = |x| - x$

$f(x) = \begin{cases} x - x & \text{when } x \geq 0 \\ -x - x & \text{when } x < 0 \end{cases}$

$= \begin{cases} 0 & \text{when } x \geq 0 \\ -2x & \text{" } x < 0 \end{cases}$

Domain: $D_f = [0, \infty) \cup (-\infty, 0)$

$= (-\infty, \infty) = \mathbb{R}$

Range:- For, $[0, \infty)$, $f(x) = 0$, $R_f = \{0\}$

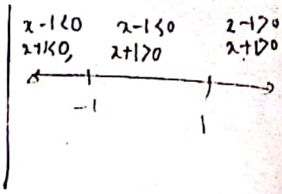
For, $(-\infty, 0)$, $f(x) = -2x$, $R_f = (0, \infty)$

Range of this f^n, $R_f = \{0\} \cup (0, \infty) = [0, \infty)$.

#/ Find domain and Range of $f(x) = |x+1| + |x-1|$

Soln:- Given, $f(x) = |x+1| + |x-1|$

$$\therefore f(x) = \begin{cases} (x+1) + (x-1) & \text{when } x \geq 1 \\ (x+1) - (x-1) & \text{when } -1 \leq x < 1 \\ -(x+1) - (x-1) & \text{when } x < -1 \end{cases}$$



$$= \begin{cases} 2x & \text{when } x \geq 1 \\ 2 & \text{when } -1 \leq x \leq 1 \\ -2x & \text{when } x < -1 \end{cases}$$

Domain of this f^n, $D_f = [1,\infty) \cup [-1,1) \cup (-\infty,-1)$

$= (-\infty,\infty) = \mathbb{R}$

Range: For, $[1,\infty)$; $f(x) = 2x$, $R_f = [2,\infty)$

For, $[-1,1)$; $f(x) = 2$, $R_f = \{2\}$

For, $[-1,-\infty)$; $f(x) = -2x$; $R_f = (2,\infty)$

$\therefore$ Range of this f^n; $R_f = [2,\infty) \cup \{2\} \cup (2,\infty)$

$= [2,\infty)$

$f(x) = |x| + |x-1|$

Soln:-

$$f(x) = |x| + |x-1|$$

$$= \begin{cases} -x-(x-1) & \text{when } x<0 \\ x-(x-1) & \text{when } 0 \le x < 1 \\ x+x-1 & \text{when } x \ge 1 \end{cases}$$

$$= \begin{cases} -2x+1 & \text{when } x<0 \\ 1 & ; \ 0 \le x < 1 \\ 2x-1 & ; \text{ when } x \ge 1 \end{cases}$$

Domain, $D_f = (-\infty, 0) \cup [0, 1) \cup [1, \infty)$

$= (-\infty, \infty) = \mathbb{R}$.

$R_f = [1, \infty) \cup \{1\} \cup [1, \infty)$

$= [1, \infty)$.

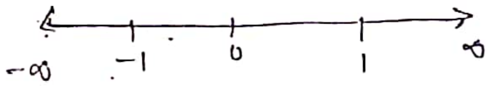
$f(x) = |x-1| - |x-2|$ → H.W.

Hints: $D_f = \mathbb{R}$.

$R_f = [-1, 1]$.

$f(x) = |x+1| + |x| + |x-1|$

Soln:



$$= \begin{cases} -(x+1) - x - (x-1) & ; \ x < -1 \\ x+1 - x - (x-1) & ; \ -1 \le x < 0 \\ (x+1) + x - (x-1) & ; \ 0 \le x < 1 \\ x+1 + x + x - 1 & ; \ x \geq 1 \end{cases}$$

$$= \begin{cases} -3x & ; \ x < -1 \\ -x+2 & ; \ -1 < x < 0 \\ x+2 & ; \ 0 \le x < 1 \\ 3x & ; \ x \geq 1 \end{cases}$$

Domain, $D_f = (-\infty, -1) \cup [-1, 0) \cup [0, 1) \cup [1, \infty)$

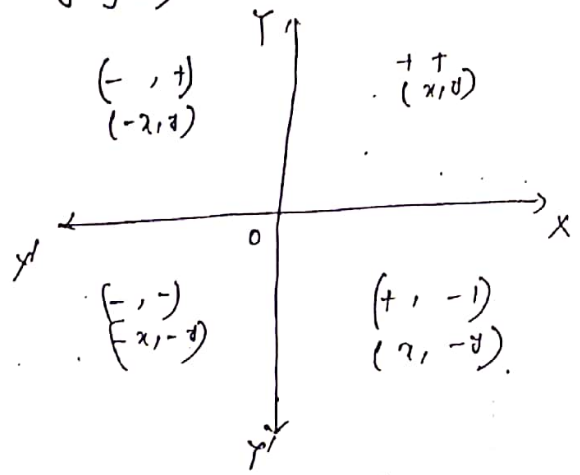
$= (-\infty, \infty) = \mathbb{R}$

$R_f = [3, \infty) \cup (2, 3] \cup [2, 3) \cup [3, \infty)$

$= [2, \infty)$

Graph, Domain and Range of a function

Graph of a function:- The graph of a function $y=f(x)$ is the locus of all points (x,y) satisfied by the equation $y=f(x)$.



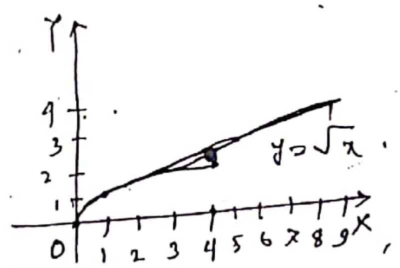
Graph of Some common f^n:

(i) $y = f(x) = \sqrt{x}$

Sol^n:

$y = \sqrt{x}$

x	0	1	4	9
y	0	1	2	3



(ii) $y = \frac{1}{x}$

$\therefore y = f(x) = \frac{1}{x}$

$\therefore f(0) = \frac{1}{0}$ undefined.

$y = \frac{1}{x};\ x<0$

x	-2	-1	$-1/2$	$-1/3$
y	$-1/2$	-1	-2	-3

$y = \frac{1}{x};\ x>0$

x	$1/3$	$1/2$	1	2
y	3	2	1	$1/2$

Find the domain and range of

$$f(x) = \begin{cases} 5 & ; 0 < x < 1 \\ 10 & ; 1 < x \le 2 \\ 15 & ; 2 < x \le 3 \end{cases}$$

$D_f = (0, 1) \cup (1, 2] \cup (2, 3]$.

$= (0, 3] - \{1\}$.

Range: $R_f = \{5, 10, 15\}$.

Assignment

Draw the graphs of the functions, and find their domains and ranges

(i) $f(x) = -x^2+1$; $D_f = \mathbb{R}$, $R_f : y \le 1$

(ii) $f(x) = \begin{cases} x-1 & ; 0 < x < 1 \\ 2x & ; 1 \le x \end{cases}$ $\rightarrow$ $D_f : x > 0$ & $R_f : -1 < y < 0$ or $y \ge 2$

(iii) $y = f(x) = \dfrac{x^2-4}{x-2}$; $D_f : x \ne 2$, $R_f : y \ne 4$

(iv) $y = f(x) = \dfrac{1}{x}$; $D_f : x \ne 0$, $R_f : y \ne 0$

(v) $y = f(x) = x - |x|$; $D_f = \mathbb{R}$, $R_f : y \le 0$..

(vi) $f(x) = \sqrt{\dfrac{x+1}{x-1}}$; $D_f = (-\infty, -1) \cup (1, \infty)$

& $R_f = \mathbb{R} - \{-1, 1\}$.

Draw the graph of the function defined by

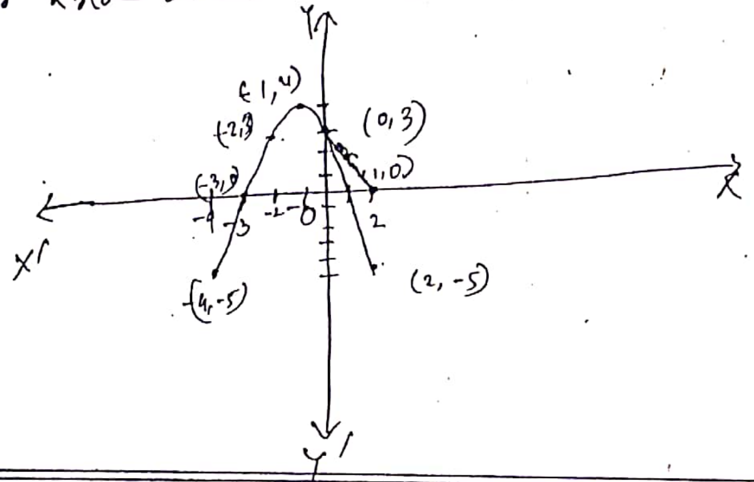
$y = 3 - 2x - x^2$.

Soln:- let $y = f(x) = 3 - 2x - x^2$

Now we draw the following table.

x	-5	-4	-3	-2	-1	0	1	2	3	4
y	-12	-5	0	3	4	3	0	-5	-12	-21

From the table we plot the point in the graph papers and draw the graph through this point.



let $y = f(x) = \begin{cases} x^2+1 & \text{when } x<0 \\ x & \text{when } 0 \le x \le 1 \\ 1/x & \text{when } 1<x \end{cases}$

Soln: Now we draw the following tables.

(i) $y = x^2+1$; $x<0$

x	-1	$-1/2$	$-1/4$
y	2	$5/4$	$17/16$

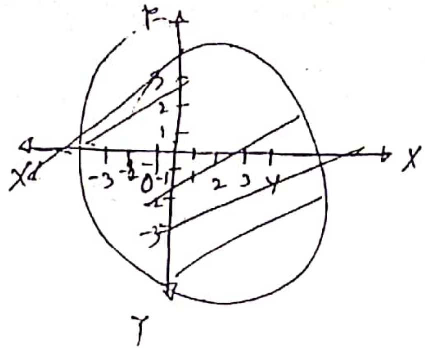
(ii) $y = x$; $0 \le x \le 1$

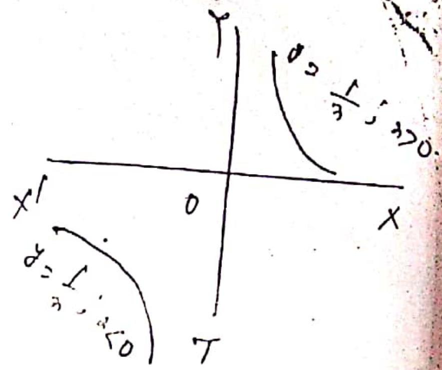
x	0	$1/2$	1
y	0	$1/2$	1

(iii)

x	$4/3$	2	3
y	$3/4$	$1/2$	$1/3$

From the table we plot the point in the graph papers and draw the graph through this point.



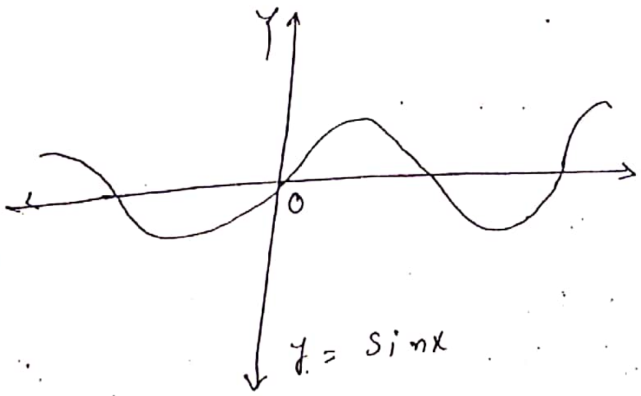


$y = \sin x$

Now we draw the following table

x	$-\pi$	$-\pi/2$	0	$\pi/2$	π	$3\pi/2$	2π
y	0	-1	0	1	0	-1	0

from the table we plot the point in the graph paper and draw the graph through this point.

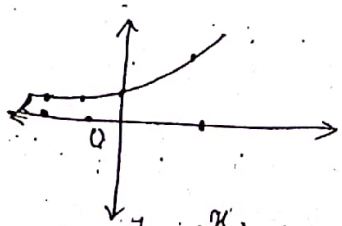


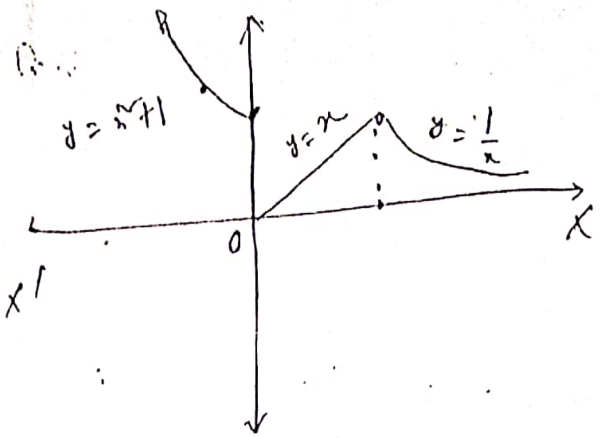
Here, $D_f = \mathbb{R}$ & $R_f = [-1, 1]$

Similarly, cosx, tanx

$y = f(x) = a^x$; $a>1$.

x	-3	-2	-1	0	1	2
y	$1/8$	$1/4$	$1/2$	1	2	4





Here, $D_f = (-\infty, 0) \cup [0, 1] \cup (1, \infty)$

$= (-\infty, \infty)$

$R_f = [1, \infty) \cup [0, 1] \cup (0, 1)$

$= [0, \infty)$

7/ $y = |x| + |x-1|$ [Previous problem]

Soln:

Given, $y = f(x) = |x| + |x-1|$

$$= \begin{cases} -2x+1 & ; x < 0 \\ 1 & ; 0 \leq x < 1 \\ 2x-1 & ; x \geq 1 \end{cases}$$

Now, we draw the following table:

(i) $y = -2x+1;\ x<0$

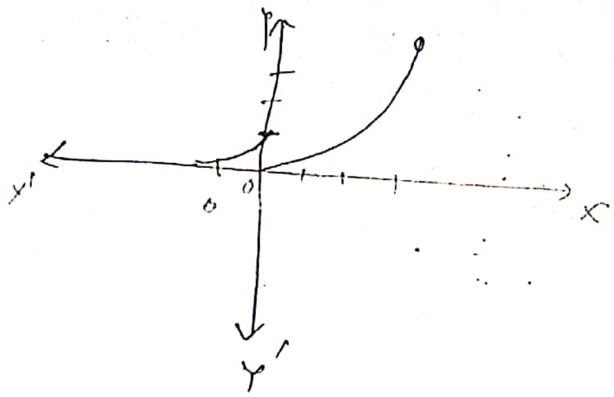
x	-2	-1	$-1/10$
y	5	3	$6/5$

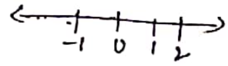
(ii) $y = 1;\ 0 \leq x < 1$

x	0	$.9$
y	1	1

(iii) $y = 2x-1;\ 1 \leq x$

x	1	2	3
y	1	3	5





Here, $D_f = (-1,0) \cup [0,2)$
$= (-1, 2)$.

and $R_f = (e^{-1/2}, 1) \cup \boxed{[0,4)}$
$= [0, 4)$.

Assignment :- Draw the graph and find domain & Range.

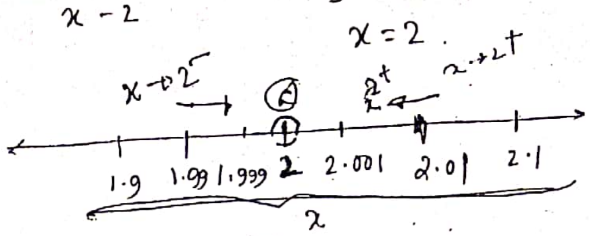
(i) $y = f(x) = \begin{cases} 3+2x & ; \ -3/2 < x < 0 \\ 3-2x & ; \ 0 \le x < 3/2 \\ 3+2x & ; \ 3/2 \le x . \end{cases}$

(ii) $f(x) = \begin{cases} 0 & ; \ |x| > 1 \\ 1+x & ; \ -1 \le x \le 0 \\ 1-x & ; \ 0 < x \le 1 . \end{cases}$

1F(5)

(2)

* $f(x) = \frac{x^2 - 4}{x - 2}$ this function is undefined when $x = 2$.



$x = 1.99$, $f(x) = 3.9$
$x = 1.99$, $= 3.99$
$x = 1.999$, $= 3.999$

$x = 2.1$, $f(x) = 4.1$
$x = 2.01$, $f(x) = 4.01$
$x = 2.001$, $f(x) = 4.001$

$$f(1.999) = \frac{(1.999)^2 - 4}{1.999 - 2} = 3.999 \text{ (L.H.L)}$$

$$f(2.001) = \frac{(2.001)^2 - 4}{2.001 - 2}$$

$$= 4.001 \text{ (R.H.L)}$$

$$f(x) = \frac{x^2 - 2^2}{x - 2}$$

$$= \frac{(x+2)\cancel{(x-2)}}{\cancel{(x-2)}}$$

$$= 2 + 2 \quad [x = 2]$$

$$= 4 \text{ (L)}$$

* $x \to a$ (x এর মান ক্রমশ a এর দিকে ধাবিত হচ্ছে।

* $|x - a| \to$ [illegible] মান [illegible] এর [illegible] [illegible] [illegible] [illegible]।

* a [illegible] x এর [illegible] L.H.L.

* L.H.L (Left hand limit)
$x \to a^-$

R.H.L (Right hand limit)

R.H.L

Limit of a function:-

When the variable x is approaches the fixed the number 'a' from both sides ultimately becomes very near equal to 'a'. if $f(x)$ approaches a fixed number 'l'. Then 'l' is said to be the limit of $f(x)$.

This is expressed by writting $\lim_{x \to a} f(x) = l$

$\lim_{x \to a^+} f(x)$ is called Right hand limit

$\lim_{x \to a^-} f(x)$ is called left hand limit.

P-1: Given,

$$f(x) = \frac{x^2 - 5x + 6}{x - 3}; \text{ does } \lim_{x \to 3} f(x) \text{ exist?}$$

Soln:- Given,

$$f(x) = \frac{x^2 - 5x + 6}{x - 3}$$

$$= \frac{(x-3)(x-2)}{x-3}$$

$$= x - 2$$

L.H.L.,

$$\lim_{x \to 3^-} f(x)$$

$$= \lim_{x \to 3} (x-2)$$

$$= 3 - 2$$

$$= 1.$$

$$R.H.L = \lim_{x \to 3^+} f(x)$$

$$= \lim_{x \to 3^+} (x-2)$$

$$= 3 - 2$$

$$= 1.$$

Since L.H.L = R.H.L ; Therefore $\lim_{x \to 3} f(x)$ exist.

---

(2) H.W   Given $f(x) = \dfrac{x^2-25}{x-5}$ does $\lim_{x \to 5} f(x)$ exist

---

(3) If $f(x) = \begin{cases} 1+2x & \text{when } -\frac{1}{2} \le x < 0 \\ 1-2x & \text{when } 0 \le x < \frac{1}{2} \\ -1+2x & \text{when } x > \frac{1}{2} \end{cases}$

does $\lim_{x \to 0} f(x)$ and $\lim_{x \to \frac{1}{2}} f(x)$ exist?

Soln:- Given,

$$f(x) = \begin{cases} 1+2x & \text{when } -\frac{1}{2} \le x < 0 \\ 1-2x & \text{when } 0 \le x < \frac{1}{2} \\ -1+2x & \text{when } x > \frac{1}{2} \end{cases}$$

$$L.H.L = \lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (1+2x) = 1+0 = 1$$

$$R.H.L = \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (1-2x) = 1-0 = 1$$

Since L.H.L = R.H.L, therefore $\lim_{x \to 0} f(x)$ exist.

Again,

$$L.H.L = \lim_{x \to \frac{1}{2}^-} f(x) = \lim_{x \to \frac{1}{2}^-} (1-2x) = 1 - \cancel{2} \cdot \frac{1}{\cancel{2}} = 0$$

$$R.H.L = \lim_{x \to \frac{1}{2}^+} f(x) = \lim_{x \to \frac{1}{2}^+} (-1+2x) = -1 + \cancel{2} \cdot \frac{1}{\cancel{2}} = 0$$

Since L.H.L = R.H.L, therefore $\lim_{x \to \frac{1}{2}} f(x)$ exist.

④ HW Let, the function $f(x)$ is defined as follows: $f(x) = \begin{cases} x^2 & \text{when } x<1 \\ 3 & \text{when } x=1 \\ x^2+1 & \text{when } x>1 \end{cases}$

Does $\lim_{x\to 1} f(x)$ exist?

Soln: Given, $f(x) = \begin{cases} x^2 & \text{when } x<1 \\ 3 & \text{when } x=1 \\ x^2+1 & \text{when } x>1 \end{cases}$

$$L.H.L = \lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} x^2 = 1^2 = 1$$

$$R.H.L = \lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} x^2+1 = 1^2+1 = 2$$

Since $L.H.L \neq R.H.L$,

therefor $\lim_{x\to 1} f(x)$ doesn't exist.

---

⑤ Let, $f(x) = \dfrac{3x+|x|}{7x-5|x|}$ does $\lim_{x\to 0} f(x)$ exist?

Soln: Given $f(x) = \dfrac{3x+|x|}{7x-5|x|}$

$$L.H.L = \lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{3x-x}{7x+5x} = \frac{2x}{12x} = \frac{1}{6}.$$

$$= \lim_{x\to 0^-} \frac{3x+|x|}{7x-5|x|}$$

R.H.L., $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \dfrac{3x+|x|}{7x-5|x|}$

$$= \frac{3x+x}{7x-5x}$$

$$= \frac{4x}{2x} = 2$$

Since, L.H.L $\neq$ R.H.L, therefore, $\lim_{x \to 0} f(x)$ doesn't exist.

---

(6) Given, $f(x) = \begin{cases} x & \text{when } x>0 \\ 0 & \text{when } x=0 \\ -x & \text{when } x<0 \end{cases}$

H.W

does $\lim_{x \to 0} f(x)$ exist?

Soln:- Given, $f(x) = \begin{cases} x & \text{when } x>0 \\ 0 & \text{when } x=0 \\ -x & \text{when } x<0 \end{cases}$

$L.H.L = \lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} -x = 0$

$R.H.S, = \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} x = 0.$

Since, L.H.L = R.H.L.

therefore, $\lim_{x \to 0} f(x)$ exist.

Continuity of a function

Continuity:- A function $f(x)$ is said to be continuous for $x = a$, provided $\lim_{x \to a} f(x)$ exists, is finite, and is equal to $f(a)$

In other words, for $f(x)$ to be continuous at $x =$

$$\lim_{x \to a+0} f(x) = \lim_{x \to a-0} f(x) = f(a)$$

Ex:- $f(x) = x^2$ is continuous for any value a of x for, $\lim_{x \to a} x^2 = a^2$.

Discontinuity of a function

A function which is not continuous at point is said to have a discontinuity at that point.

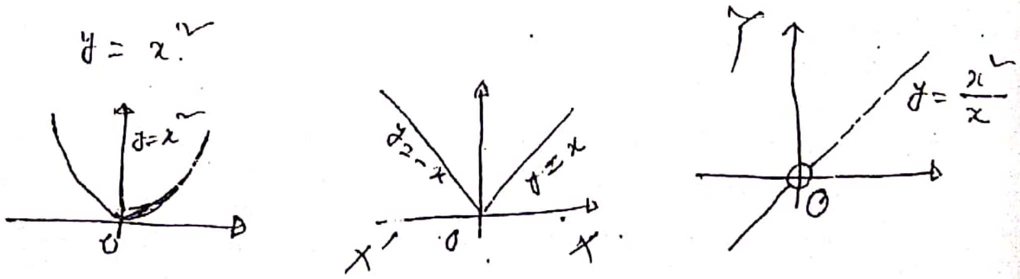
Ex:- $f(x) = \frac{1}{x^2}$ is discontinuous at $x = 0$, for $\lim_{x \to 0} \left(\frac{1}{x^2}\right)$ is not finite.

or,

$f(x) = \cos\frac{1}{x}$ is discontinuous at $x = 0$, since $\lim_{x \to 0} \cos\frac{1}{x}$ does not exist.

Ex:-

$y = x^2$



Test the continuity of the following functi.
at $x=2$, $f(x) = \begin{cases} 5 & \text{when } 0<x<1 \\ 10 & \text{when } 1<x\le 2 \\ 15 & \text{when } 2<x\le 3. \end{cases}$

Soln:- When $x=2$

$\Rightarrow f(x) = 10$

$\Rightarrow f(2) = 10$

Soln

$L.H.L = \lim_{x\to 2^-} f(x)$

$= \lim_{x\to 2^-} 10$

$= 10$

$R.H.L = \lim_{x\to 2^+} f(x)$

$= \lim_{x\to 2^+} 15$

$= 15$

Since $L.H.L \neq R.H.L$, therefore $\lim_{x\to 2} f(x)$ doesn't exist.

The function is not continuous at point $x=2$.

Test the continuity of the followi function at $x=0$ and $x=\frac{3}{2}$,

$$f(x) = \begin{cases} 3+2x & \text{when } -\frac{3}{2}\le x<0 \\ 3-2x & \text{when } 0\le x<\frac{3}{2} \\ -3-2x & \text{when } x\geq \frac{3}{2} \end{cases}$$

4

Soln:-

Given, $f(x) = \begin{cases} 3+2x & \text{when } -\frac{3}{2} \le x < 0 \\ 3-2x & \text{''} \quad 0 \le x < \frac{3}{2} \\ -3-2x & \text{''} \quad x \geq \frac{3}{2} \end{cases}$

When,

$x=0$, $f(x) = 3-2x$

$f(0) = 3-0 = 3$

$$\text{L.H.L} = \lim_{x\to 0^-} f(x) = \lim_{x\to 0^-}(3+2x) = 3+0 = 3$$

$$\text{R.H.L} = \lim_{x\to 0^+} f(x) = \lim_{x\to 0^+}(3-2x) = 3-0 = 3.$$

Since, L.H.L = R.H.L, therefore, $\lim_{x\to 0} f(x)$ does exist.

Again

when, $x = \frac{3}{2}$; $f(x) = -3-2x$

$\Rightarrow f(3/2) = -3-2\cdot\frac{3}{2}$

$= -6.$

$$\text{L.H.L} = \lim_{x\to \frac{3}{2}^-} f(x) = \lim_{x\to\frac{3}{2}^-}(3-2x) = 3-2\cdot\frac{3}{2} = 0.$$

$$\text{R.H.L} = \lim_{x\to\frac{3}{2}^+} f(x) = \lim_{x\to\frac{3}{2}^+}(-3-2x) = -3-2\cdot\frac{3}{2} = -6.$$

the

ing

Since, L.H.L $\neq$ R.H.L, therefore $\lim_{x \to \frac{3}{2}} f(x)$ doesn't exist.

So the function is not continuous at the point $x = \frac{3}{2}$ but continuous at $x = 0$.

Assignment

(8) Test the continuity at $x = 0$, $x = 1$

$$f(x) = \begin{cases} x^2+1 & \text{when } x < 0 \\ x & \text{when } 0 \le x \le 1 \\ 1/x & \text{when } x > 1 \end{cases}$$

---

(9) Test the continuity at $x = 0, 1, 2$

Assignment.

$$f(x) = \begin{cases} -x^2 & \text{when } x \le 0 \\ 5x-4 & \text{when } 0 < x \le 1 \\ 4x^2-3x & \text{when } 1 < x < 2 \\ 3x+4 & \text{when } x \ge 2 \end{cases}$$

---

(5) Test the continuity at $x = 0$

$$f(x) = \begin{cases} \frac{|x|}{x} & \text{when } x \neq 0 \\ 1 & \text{when } x = 0. \end{cases}$$

Soln:- When, $x = 0$, $f(x) = 1 \Rightarrow f(0) = 1$

$$L.H.L = \lim_{x \to 0^-} f(x)$$

$$= \lim_{x \to 0^-} \frac{|x|}{x}$$

$$= \frac{-x}{x}$$

$$= -1$$

$$R.H.L = \lim_{x \to 0^+} f(x)$$

$$= \lim_{x \to 0^+} \frac{|x|}{x}$$

$$= \frac{x}{x} = 1$$

Since $L.H.L \neq R.H.L$, therefore $\lim_{x \to 0}$ doesn't exist.

So the function is not continuous at $x = 0$.

---

(6) Test the continuity at $x=0$

$$f(x) = \begin{cases} -\cos x & \text{when } x \neq 0 \\ \cos x & \text{when } x = 0. \end{cases}$$

(Q7) Test the continuity at $x=0$.

$$f(x) = \begin{cases} \frac{\log_e (1+x)}{\tan^{-1} x} & \text{when } x \neq 0 \\ 1 & \text{when } x = 0. \end{cases}$$

Soln:- Given $f(x) =$

When $x = 0$, $f(0) = 1$

$$L.H.L. = \lim_{x \to 0^-} f(x)$$

$$= \lim_{x \to 0^-} \frac{\log_e(1+x)}{\tan^{-1} x}$$

$$= \lim_{x \to 0^-} \frac{\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots\right)}{\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots\right)}$$

$$= \lim_{x \to 0^-} \frac{x\left(1 - \frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \cdots\right)}{x\left(1 - \frac{x^2}{3} + \frac{x^4}{5} - \cdots\right)}$$

$$= \lim_{x \to 0^-} \frac{1 - \frac{x}{2} + \frac{x^2}{3} - \cdots}{1 - \frac{x^2}{3} + \frac{x^4}{5} - \cdots}$$

$$= \frac{1}{1} = 1$$

$$R.H.L. = \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \frac{\log_e(1+x)}{\tan^{-1} x} = 1$$

$\therefore$ $\lim_{x \to 0} f(x)$ exist.

Since $\lim_{x \to 0} f(x) = f(0)$; therefore the function is continuous at $x = 0$.

* Test the continuity at $x=1,2$

$$f(x) = \begin{cases} \log x & \text{when } 0 < x \le 1 \\ 0 & \text{when } 1 < x < 2 \\ 1+x^2 & \text{when } x > 2 \end{cases}$$

When,

$x = 1$

$f(x) = \log x$

$\Rightarrow f(1) = \log(1) = 0$

L.H.L $= \lim_{x\to 1^-} f(x)$

$= \lim_{x\to 1^-} \log x$

$= \log(1)$

$= 0$

R.H.L,

$\lim_{x\to 1^+} f(x)$

$= \lim_{x\to 1^+} 0$

$= 0$

$\therefore \lim_{x\to 1} f(x)$ exist.

Since $\lim_{x\to 1} f(x) = f(1)$ therefore the function is Continuous at $x=1$.

When $x=2$, $f(x)=0$, $\Rightarrow f(2)=0$.

L.H.L, $= \lim_{x\to 2^-} f(x)$

$= \lim_{x\to 2^-} 0 = 0$

R.H.L, $= \lim_{x\to 2^+} f(x)$

$= \lim_{x\to 2^+} (1+x^2)$

$= 1+(2)^2$

$= 1+4 = 5$

Since, L.H.L $\neq$ R.H.L

So the $\lim_{x\to 2} f(x)$ does not exist.

Therefore the function is not continuous at $x=2$.

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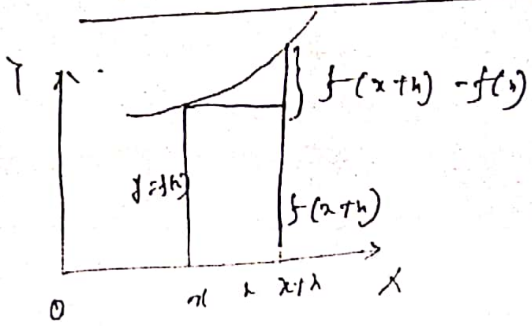
Differentiability of a f^n



$$\tan\theta = \frac{dy}{dx} = \frac{f(x+h)-f(x)}{x+h-x} = \frac{f(x+h)-f(x)}{h}$$

If $h \to 0$

$$\frac{dy}{dx} \text{ or } y' \text{ or } f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

Condition:

Let $f(x)$ be a function which is defined in the interval $[a,b] \subset R$ and let $a<c<b$ then the function $f(x)$ is said to be differentiable at $x=c$ if $f'(c) = \lim_{h\to 0} \frac{f(c+h)-f(c)}{h}$ exist?

and $Rf'(c) = \lim_{h\to 0^+} \frac{f(c+h)-f(c)}{h}$

and $Lf'(c) = \lim_{h\to 0^-} \frac{f(c\cancel{+}h)-f(c)}{h}$ are finite and equal $[Lf'(c) = Rf'(c)]$

• Test whether the f^n $f(x) = x^2+5$ is differentiable at the point $x=1$.

$$Rf'(c) = \lim_{h \to 0^+} \frac{f(c+h) - f(c)}{h}$$

$f(1) = 1+5 = 6$

$f(1+h) = (1+h)^2+5$

$$\therefore Rf'(1) = \lim_{h \to 0^+} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \to 0^+} \frac{(1+h)^2+5-6}{h}$$

$$= \lim_{h \to 0^+} \frac{1+2h+h^2+5-6}{h}$$

$$= \lim_{h \to 0^+} \frac{2h+h^2}{h}$$

$$= \lim_{h \to 0^+} \frac{\cancel{h}(2+h)}{\cancel{h}}$$

$$= 2+0 = 2.$$

$$Lf'(c) = \lim_{h \to 0^-} \frac{f(c+h) - f(c)}{h}$$

$$Lf'(1) = \lim_{h \to 0^-} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \to 0^-} \frac{1+2h+h^2+5-6}{h}$$

$$= \lim_{h \to 0^-} (2+h) = 2+0 = 2$$

Since $Lf'(1) = Rf'(1)$ and both are finite so the f^n is differentiable at $x=1$

A f^n $f(x)$ is defined as

$$f(x) = \begin{cases} 5x+4 & \text{when } 0<x\le 1 \\ 4x^2-3x & \text{when } 1<x<2 \end{cases}$$

(i) Test the continuity at $x=1$ — H.W

(ii) Test whether the above f^n is differentiable at $x=1$

Soln:- $Rf'(1) = \lim_{h\to 0^+} \frac{f(1+h)-f(1)}{h}$

$= \lim_{h\to 0^+} \frac{4(1+h)^2-3(1+h)-1}{h}$

$= \lim_{h\to 0^+} \frac{4(1+2h+h^2)-3-3h-1}{h}$

$= \lim_{h\to 0^+} \frac{4+8h+4h^2-3-3h-1}{h}$

$= \lim_{h\to 0^+} \frac{1+5h+4h^2-1}{h}$

$= \lim_{h\to 0^+} \frac{\cancel{h}(4h+5)}{\cancel{h}}$

$= \lim_{h\to 0^+} (4h+5)$

$= 5$

If $x>1$
$\Rightarrow x-1>0$
$\Rightarrow h>0$ i.e, $h\to 0^+$
$f(x) = 4x^2-3x$
$f(1) = 4-3 = 1$
$f(1+h) = 4(1+h)^2-3(1+h)$
$= 4(1+2h+h^2)-3-3h$
$= 4+8h+4h^2-3-3h$
$= 1+5h+4h^2$

$Lf'(1) = \lim_{h\to 0^-} \frac{f(1+h)-f(1)}{h}$

$= \lim_{h\to 0^-} \frac{\cancel{9}+5h-\cancel{9}}{h}$

$= \lim_{h\to 0^-} (5)$

$= 5$

If $x<1$, $f(x) = 5x+4$
$x-1<0$
$h<0 \Rightarrow h\to 0^-$
$f(1) = 5+4 = 9$
$f(1+h) = 5(1+h)+4$
$= 5+5h+4$
$= 5h+9$

Since $Rf'(1) = Lf'(1)$ and both are finite
So the f^n is differentiable at $x=1$.

More problems: From recommended books.

$\therefore R.H.L = \lim_{x \to \frac{\pi}{2}+} f(x) = \lim_{x \to \frac{\pi}{2}+} 2 + (x - \frac{\pi}{2})^2$

$= \lim_{x \to \frac{\pi}{2}} 2 + (\frac{\pi}{2} - \frac{\pi}{2})^2 = 2 + 0 = 2$

~~L.H~~ . When $x < \frac{\pi}{2}$ then $f(x) = 1 + \sin x$

$L.H.L = \lim_{x \to \frac{\pi}{2}-} f(x) = \lim_{x \to \frac{\pi}{2}-} \cancel{f(x)}^{1+\sin x} = 1 + \sin \frac{\pi}{2} = 1$

When $x = \frac{\pi}{2}$ then $f(x) = 2 + (x$

$f(\frac{\pi}{2}) = 2 + (\frac{\pi}{2} - \frac{\pi}{2})^2 = 2$

Since, $\lim_{x \to \frac{\pi}{2}+} f(x) = \lim_{x \to \frac{\pi}{2}-} f(x) = f(\frac{\pi}{2}$

Therefore, $x = \frac{\pi}{2}$ $f(x)$ is continuous.

The fn $f(x)$ is continuous at $x = \pi/2$

For differentiability:

$Rf'(x) = \lim_{h \to 0+} \frac{f(x+h) - f(x)}{h}$

$\Rightarrow Rf'(\pi/2) = \lim_{h \to 0+} \frac{f(\pi/2 + h) - f(\pi/2)}{h}$

$= \lim_{h \to 0+} \frac{\cancel{2} + (\frac{\pi}{2} + h - \frac{\pi}{2})^2 - \cancel{2}}{h}$

$= \lim_{h \to 0+} \frac{h^2}{h} = \lim_{h \to 0+} h = 0$

When $x < \frac{\pi}{2}$ then $f(x) = 1 + \sin x$

$\lim$ ... $f(\frac{\pi}{2} + h) - f(\frac{\pi}{2})$ ... $1 + \sin(\frac{\pi}{2}$

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$$= \lim_{h \to 0^-} \frac{\cos h - 1}{h} = \lim_{h \to 0^-} \frac{\left(1 - \frac{h^2}{2!} + \frac{h^4}{4!} - \cdots\right) - 1}{h} \quad \left[\cos x = 1 - \frac{x^2}{2!} \ldots\right]$$

$$= \lim_{h \to 0^-} \frac{-\frac{h^2}{2!} + \frac{h^4}{4!} - \cdots}{h}$$

$$= \lim_{h \to 0} (\ldots) = 0$$

$\because$ $Lf'(\frac{\pi}{2}) = Rf'(\frac{\pi}{2})$ therefor the f^n $f(x)$ is differentiable

* (ϵ, δ) defn of limit : $\lim_{x \to a} f(x) = l$. Given a number $\epsilon > 0$, however small, there exists ($\exists$) a positive number δ such that for x ($\forall$) on a $\delta > 0$ satisfying $|x - a| < \epsilon$

$\Rightarrow |f(x) - l| < \epsilon$

1. prove that $\rightarrow \lim_{x \to 2} \underset{f(x)}{(2x+1)} = \underset{l}{5}$

$\because$ Let us consider an arbitary positive number $\epsilon > 0$, however s

then $|f(x) - 5| = |2x + 1 - 5| < \epsilon \Rightarrow |2x - 4| < \epsilon$

$= |2(x-2)| < \epsilon$

$\Rightarrow |f(x) - 5| <$

i.e., $|x - 2| < \frac{\epsilon}{2}$

$\Rightarrow |x - 2| < \delta \quad | \quad \delta = \frac{\epsilon}{2} > 0$

~~lim~~ this means that $\lim_{x \to 2} (2x+1) = 5$

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A f^n is defined as $f(x) = \begin{cases} 5x-4 & \text{when } 0<x\leq 1 \\ 4x^2-3x & \text{"} \quad 1<x<2 \end{cases}$

(I) Does limit exist $x \to 9$

(II) Test the continuity at $x=1$

(III) Test whether the above f^n is different at $x=a$.

(II) When $x \neq 1$ then $f(x) = 5x-4$

$$L.H.L = \lim_{\substack{x\to 1^- \\ x>0}} f(x) = \lim_{x\to 1^-} (5x-4) = 5\cdot 1-4=1$$

$$R.H.L = \lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} (4x^2-3x) = 4\cdot 1^2 - 3\cdot 1 = 1$$

When $x=1$

$$f(x) = 5x-4$$
$$= 5\cdot 1 - 4$$
$$= 5$$

Since, $L.H.L = R.H.L$, therefore $\lim_{x\to 0} f(x)$ does

$$f'(x)_{x=c} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \qquad Lf'(c) =$$

if, $Rf'(c) = \lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h} = \lim_{h\to 0^-} \frac{f(a-h)-f(a)}{h}$

The f^n $f(x)$ is differentiable a

$$Lf'(c) = Rf'(c)$$

Sol^n:-

$$Rf'(1) = \lim_{h\to 0^+} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h\to 0^+} \frac{4(1+h)^2 - 3(1+h) - (4.1^2 - 3.1)}{h}$$

$$= \lim_{h\to 0^+} \frac{h(4h+5)}{h}$$

$$= 5.$$

of $x>1$, $x-1>0$, $h>0$, i.e. h: $f+1$

$$Lf'(1) = \lim_{h\to 0^-} \frac{f(1+h) - f(1)}{h}$$

$$5(1+h)-4-1$$

$$= \lim_{h\to 0^-} \frac{\cancel{9+5h-9}}{h}$$

$h<1$, $x-1<0$, $h<0$, $h\to 0$, $f-(1)$

$$= \lim_{h\to 0} \frac{\cancel{5}+5h-\cancel{5}}{h}$$

$$= 5$$

(2) A f^n $f(x) = \begin{cases} 1 & ; x<0 \\ 1+\sin x & ; 0 \le x < \frac{\pi}{2} \\ 2+\left(x-\frac{\pi}{2}\right)^2 & ; x \geq \frac{\pi}{2} \end{cases}$

Investigate the continuity and differentiability of the f^n $f(x)$ at $x = \frac{\pi}{2}$

Sol^n: when $x > \frac{\pi}{2}$ then $f(x) = 2 + \left(x-\frac{\pi}{2}\right)^2$

Indeterminate form.

$\ln 0 = -\infty$, $\ln \infty = \infty$, $\ln 1 = 0$, $\ln e = 1$, $e^{\infty} = \infty$, $e^{-\infty} = 0$

(i) ∞^0, $\frac{0}{0}$, $\frac{\infty}{\infty}$, $\infty \times \infty$, $\frac{a}{0}$; $\frac{1}{0}$, $\infty - \infty$, $\infty + \infty$

$\infty \times 0$, 0^0, , $0 \times \infty$,

(ii) Note: $1^{+\infty} = 1$, $\frac{1}{\infty} = 0$, $\frac{a}{\infty} = 0$.

Then may be appliable of L Hospital's theorem

$(1+x)^{-1} = 1 + x + x^2 + x^3 + \cdots$

$(1+x)^{-1} = 1 - x - x^2 - x^3$

Important Expansion Series.

$(1+x)^n$

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) \cdots \infty$$

(i) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \infty$

(ii) $e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots \infty$

(iii) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \infty$

(iv) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \infty$

(v) $\log(1+x) = x - \frac{x^2}{20} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \infty$

(vi) $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \cdots \infty$

(vii) $\tan x = x + \frac{x^3}{3} + \frac{2}{15} x^5 + \cdots \infty$

(viii) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \infty$

* (i) $\lim_{x \to 0} \frac{\sin x}{x} = 1$ (ii) $\lim_{x \to 0} \frac{x}{\sin x} = 1$ (v) $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$

(iii) $\lim_{x \to 0} \frac{\tan x}{x} = 1$ (iv) $\lim_{x \to 0} \frac{x}{\tan x} = 1$ (vi) $\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1$

(iv) $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \cdots$

$(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 \cdots$

1. $\lim_{x\to 0} \dfrac{xe^x - \log(1+x)}{x^2}$ $\left[\frac{0}{0} \therefore \text{using L'Hospital}\right]$

$= \lim_{x\to 0} \dfrac{(e^x + xe^x) - \frac{1}{1+x}}{2x}$ $\left[\frac{0}{0}\right]$

$= \lim_{x\to 0} \dfrac{e^x + (e^x + xe^x) + \frac{1}{(1+x)^2}}{2}$

$= \dfrac{1 + 1 + 0 + \frac{1}{(1+0)^2}}{2}$

$= \dfrac{3}{2}$ Ans.

2. $\lim_{x\to 0} \dfrac{e^x - e^{-x} + 2\sin x - 4x}{x^5}$ $\left[\frac{0}{0}\right]$

$= \lim_{x\to 0} \dfrac{e^x + e^{-x} + 2\cos x - 4}{5x^4}$ $\left[\frac{0}{0}\right]$

$= \lim_{x\to 0} \dfrac{e^x - e^{-x} - 2\sin x}{20x^3}$ $\left[\frac{0}{0}\right]$

$= \lim_{x\to 0} \dfrac{e^x + e^{-x} - 2\cos x}{60x^2}$ $\left[\frac{0}{0}\right]$

$= \lim_{x\to 0} \dfrac{e^x - e^{-x} + 2\sin x}{120x}$ $\left[\frac{0}{0}\right]$

$= \lim_{x\to 0} \dfrac{e^x + e^{-x} + 2\cos x}{120}$

$= \dfrac{1+1+2\cdot 1}{120} = \dfrac{4}{120} = \dfrac{1}{30}.$

3. $$\lim_{\theta \to \frac{\pi}{4}} \frac{\sqrt{2} - \cos\theta - \sin\theta}{(4\theta - \pi)^2}$$

$$= \lim_{\theta \to \frac{\pi}{4}} \frac{\sin\theta - \cos\theta}{2(4\theta - \pi)\cdot 4} \quad \left[\frac{0}{0}\right]$$

$$= \lim_{\theta \to \frac{\pi}{4}} \frac{\cos\theta + \sin\theta}{8\cdot 4}$$

$$= \frac{\cos\frac{\pi}{4} + \sin\frac{\pi}{4}}{32}$$

$$= \frac{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{32}$$

$$= \frac{2/\sqrt{2}}{32}$$

$$= \frac{\sqrt{2}}{32}$$

$$= \frac{1}{16\sqrt{2}}$$ Ans.

4. $$\lim_{x \to a} \frac{x^n - a^n}{x - a} \quad \left[\frac{0}{0}\right]$$

$$= \lim_{x \to a} \frac{n x^{n-1}}{1}$$

$$= n a^{n-1}$$

(5) $\lim\limits_{x\to 0} \dfrac{x - \sin x}{\tan^3 x}$

X (7)

$= \lim\limits_{x\to 0} \dfrac{x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \infty\right)}{(\tan x)^3}$ (8)

$= \lim\limits_{x\to 0} \dfrac{\cancel{x} - \cancel{x} + \frac{x^3}{3!} - \frac{x^5}{5!} + \cdots \infty}{\left(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \cdots \infty\right)^3}$

$= \dfrac{\cancel{x^3}\left(\frac{1}{3!} + \frac{x^2}{5!} + \cdots + \infty\right)}{\cancel{x}\left(1 + \frac{x^2}{3} + \frac{2}{15}x^4 + \cdots + \infty\right)}$

(5) $\lim\limits_{x\to 3} \dfrac{x^2 - 9}{x - 3}$ (9)

$= \lim\limits_{x\to 3} \dfrac{2x - 0}{1 - 0}$

$= 2 \times 3$

$= 6.$

(8) $\lim\limits_{x\to 0} \dfrac{\sqrt{1+x} - 1}{x}$

$= \lim\limits_{x\to 0} \dfrac{(\sqrt{1+x} - 1)(\sqrt{1+x} + 1)}{x(\sqrt{1+x} + 1)}$ (10)

$= \lim\limits_{x\to 0} \dfrac{\cancel{x}}{\cancel{x}(\sqrt{1+x} + 1)}$ (r)

$= \lim\limits_{x\to 0} \dfrac{1}{\sqrt{1+x} + 1} = \dfrac{1}{\sqrt{1+0} + 1} = \dfrac{1}{2}$

(7) $\lim\limits_{x\to 0} \dfrac{\sqrt{1+x}-1}{x} = \dfrac{1}{2}$

(8) $\lim\limits_{x\to a} \dfrac{x^{7/2}-a^{7/2}}{\sqrt{x}-\sqrt{a}} \quad \left[\dfrac{0}{0}\right]$

$= \lim\limits_{x\to a} \dfrac{\frac{7}{\cancel{2}}\, x^{5/2}}{\frac{1}{\cancel{2}}\, x^{-\frac{1}{2}}}$

$= 7a^3$

$x^{\frac{5}{2}+\frac{1}{2}} = x^{\frac{\cancel{6}\,3}{\cancel{2}}}$

(9) $\lim\limits_{x\to 0} \dfrac{1-\cos x}{x^2}$

$= \lim\limits_{x\to 0} \dfrac{0+\sin x}{2x}$

$= \lim\limits_{x\to 0} \dfrac{\cos x}{2}$

$= \dfrac{1}{2}$ Ans.

$\lim\limits_{x\to 0} \dfrac{1-\cos x}{x^2}$

$= \lim\limits_{x\to 0} \dfrac{1-1+\frac{x^2}{2!}-\frac{x^4}{4!}+\cdots}{x^2}$

$= \lim\limits_{x\to 0} \dfrac{\cancel{x^2}\left(\frac{1}{2!}+\frac{x^2}{4!}+\cdots\right)}{\cancel{x^2}}$

$= \dfrac{1}{2!} - 0$

$= \dfrac{1}{2}$

(10) $\lim\limits_{x\to 0} x^2 \log x^2 \quad [0\times\infty]$

$= \lim\limits_{x\to 0} \dfrac{\log x^2}{x^{-2}}$

$= \lim\limits_{x\to 0} \dfrac{\frac{2x}{x^2}}{-2x^{-3}}$

$= \lim\limits_{x\to 0} \dfrac{\frac{1}{x}}{-x^{-3}}$

$= \lim\limits_{x\to 0} \left(-\dfrac{1}{x^{-2}}\right)$

$= \lim\limits_{x\to 0} (-x^2) = 0$

(11) $\lim\limits_{x \to 0} \dfrac{\tan x - \sin x}{x^3}$

$= \lim\limits_{x \to 0} \dfrac{\sec x^2 - \cos x}{3x^2}$

$= \lim\limits_{x \to 0} \dfrac{2\sec^2 x \tan x + \sin x}{6x}$

$= \lim\limits_{x \to 0} \dfrac{2[\sec^4 x + \tan^2 x \cdot 2\sec^2 x] + \cos x}{6}$

$= \dfrac{2[1+0] + 1}{6}$

$= \dfrac{2+1}{6}$

$= \dfrac{1}{2}$

Diffuerentiation

$y = f(x)$ — (I)

$y + \partial y = f(x + \partial x)$ — (II)

(II) − (I)

$\partial y = f(x + \partial x) - f(x)$

$\Rightarrow \dfrac{\partial y}{\partial x} = \dfrac{f(x+\partial x) - f(x)}{\partial x}$

$\lim\limits_{\partial x \to 0} \dfrac{\partial y}{\partial x} = \lim\limits_{\partial x \to 0} \dfrac{f(x+\partial x) - f(x)}{\partial x}$

or $\dfrac{dy}{dx} = \lim\limits_{\partial x \to 0} \dfrac{f(x+\partial x) - f(x)}{\partial x}$

[illegible]

$$= \frac{dy}{dx}, \frac{d}{dx}f(x), f'(x)$$

যদি δx কে পরিবর্তে x কে দ্বারা h ধরি

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

$$\Rightarrow x = a \rightarrow$$

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

Differential coefficient

(1) By defn, find the differential coeff of x^n.

Soln:- let, $f(x) = x^n = y$

$$\Rightarrow f(x+h) = (x+h)^n.$$

By defn, we know,

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{(x+h)^n - x^n}{h}$$

$$= \lim_{h \to 0} \frac{1}{h}\left[x^n\left(1+\frac{h}{x}\right)^n - x^n\right]$$

$$= \lim_{h \to 0} \frac{x^n}{h}\left[\left(1+\frac{h}{x}\right)^n - 1\right]$$

$$= x^n \lim_{h\to 0} \frac{1}{h}\left[\cancel{1} + n\frac{h}{x} + \frac{n(n-1)}{2!}\frac{h^2}{x^2} + \cdots \cancel{-1}\right]$$

$$= x^n \lim_{h\to 0} \frac{h}{h}\left[\frac{n}{x} + \frac{n(n-1)}{2!}\frac{h}{x^2} + \cdots\right]$$

$$\frac{d}{dx}(x^n) = x^n \cdot \frac{n}{x}$$

$$= nx^{n-1}$$

e^x

let $y = f(x) = e^x$

$f(x+h) = e^{x+h}$

By def^n $\frac{d}{dx} y = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$

$$\frac{d}{dx} e^x = \lim_{h\to 0} \frac{e^{x+h} - e^x}{h}$$

$$= \lim_{h\to 0} \frac{e^x(e^h - 1)}{h}$$

$$= e^x \lim_{h\to 0} \frac{1}{h}\left[\cancel{1} + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \cdots \cancel{-1}\right]$$

$$= e^x \lim_{h\to 0} \frac{h}{h}\left[1 + \frac{h}{2!} + \frac{h^3}{3!} + \cdots\right]$$

$$= e^x \lim_{h\to 0} \left[1 + \frac{h}{2!} + \frac{h^2}{3!} + \cdots\right]$$

$$= 1.$$

$$\frac{d}{dx} a^x = \lim_{h\to 0} \frac{a^{x+h} - a^x}{h}$$

$$= a^x \lim_{h\to 0} \frac{a^h - 1}{h}$$

$$= a^x \lim_{h\to 0} \frac{1}{h}\left[1 + \frac{h}{1!} \ln a + \frac{h^2}{2!}(\ln a)^2 + \cdots - 1\right]$$

$$= a^x \lim_{h\to 0} \left[\ln a + \frac{h}{2!}(\ln a)^2 + \cdots\right]$$

$= a^x \ln a$

H.W a^x, $\ln(x)$, $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, $\operatorname{cosec} x$

$\log_a x$, $\tan^{-1} x$, $\sin^{-1} x$, $\cot^{-1} x$, $\sec^{-1} x$, $\operatorname{cosec}^{-1} x$, $\sqrt{x}$,

$\boxed{\log_a x = \frac{1}{x}\log_a e}$, $\ln x \log_a e$

Chapter - two

ordinary differentiation

Differential Coefficients

1. principal Methods:-

General def^n of d.c.

$$y = f(x) \quad \text{--- (I)}$$

$$y + \delta y = f(x + \delta x) \quad \text{--- (II)}$$

$\{(II) - (I)\} \Rightarrow$

$$\delta y = f(x + \delta x) - f(x)$$

$$\Rightarrow \frac{\delta y}{\delta x} = \frac{f(x+\delta x) - f(x)}{\delta x}$$

$$\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta x \to 0} \frac{f(x+\delta x) - f(x)}{\delta x}$$

$$\text{or, } \frac{dy}{dx} = \lim_{\delta x \to 0} \frac{f(x+\delta x) - f(x)}{\delta x}$$

$$\text{(b)} \quad f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$x = a$

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

$\frac{dy}{dx}$

$\frac{d}{dx}$

f

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Find $\frac{dy}{dx}$ for the following function by using principal method. or By defn, find the d.c n of

1. $y = f(x) = x^n, e^x, a^x, \ln x$ or $\log_e x$, $\log_a x$

2. $y = f(x) = \sin x, \cos x, \tan x, \operatorname{cosec} x, \sec x, \cot x$

3. $y = \sin^{-1}x, \cos^{-1}x, \tan^{-1}x, \operatorname{cosec}^{-1}x, \sec^{-1}x, \cot$

MRK → $e^{\sqrt{x}}$, x^2, $\cos x$, $\sin x$, e^{5x+9}

Derive first derivatives from the followings functions.

(A) ~~$y=\sec\tan^{-1}x$~~ From H.S.C BOOK.

or,

(1) $y = \sec\tan^{-1}x$ (2) $y = \ln^2(x^2+2)$ (3) $y = 5x^{3/5} + \tan^2 2x$

(4) $\log_a x + \log_x a$ (5) $y = (\ln\sin x)^2$ (6) $y = \frac{x^n}{\log_a x}$

(7) $y = \tan^{-1}\frac{a+bx}{b-ax}$ (8) $y = \sqrt[3]{x^2+\sqrt{x}}$

(9) $y = 10^{\ln\sin x}$ (10) $y = e^{\sin x} - (\sin x^2)^2$

(11) $y = x^{x^x}$ (12) $y = x^{e^x}$ (13) $y = (\sin x)^{\ln x}$

(14) $\sin^2(\ln x^2)$ (15) $y = (\sec\tan^{-1}x)(x^5 + x^{\frac{1}{3}})^{5/3}$

(16) $y = (\ln\sin x)^2\{\log_a x + \log_x a\}$ (17) $y = \frac{x^n(x^5+x^{1/3})^{3/5}}{\log_a x}$

(18) $y = \frac{\ln(x^2+2)\ (x^5+x^{1/3})}{}$ (19) $y = \frac{x^n(x^5+x^{1/3})^{3/5}}{}$ (20) x^{e^x} ...

Successive differentiation.

Find n'th derivative for the following f's:

(i) $y = \ln x$

(ii) $y = x^m$ and x^{-m}

(iii) $y = a^x$

(iv) $y = e^{ax}$

(v) $y = \sin x$ & $\cos x$.

(vi) $y = \ln(ax+b)$

(vii) $\sin(ax+b)$ and $\cos(ax+b)$

MRK

(i) $x^2 e^{2x}$.

(ii) $y = e^x(ax$

State and prove Leibnitz's theorem:-

Prove the following:

(1.) If $y = \tan^{-1}x$ show that $(1+x^2)\,y_{n+1} + 2nxy_n + n^2 y_{n-1} - ny_{n-1} = 0$

(2.) If $y = e^{a\cos^{-1}x}$ then show that $(1-x^2)\,y_{n+2} - (2n+1)x\,y_{n+1} - (n^2+a^2)\,y_n = 0$

(3) If $y = e^{a\sin^{-1}x}$ prove that $(1-x^2)\,y_{n+2} - (2n+1)x\,y_{n+1} - (n^2+a^2)\,y_n = 0$

(4) If $x = \sin\left(\frac{1}{m} \ln y\right)$ then $(1-x^2)y_{n+2} - (2n+1)xy_{n+1}$
$-(n^2+m^2)y_n = 0$

or

If $x = \left(\frac{1}{a} \ln y\right)$ then prove that $(1-x^2)y_{n+2} - (2n$
$-(n^2+a^2)y_n = 0.$

(5) ~~Leibn~~ If $\ln y = \tan^{-1}x$, then show that
$(1+x^2)y_{n+2} + (2nx+2x-1)y_{n+1} + n(n+1)y_n = 0.$

Leibnitz's theorem: Statement:- If u & v are f^ns of x then the n'th derivative of the product of u & v is.

$(uv)_n = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2 + \cdots + {}^nc_r u_{n-r} v_r +$.

where the suffixes of u & v indicate the order of dif; w.r. to x.

Proof:- Let $y = uv$, integrating w.r. to x

$y_1 = uv_1 + u_1 v$

$y_2 = uv_2 + u_1v_1 + u_1v_1 + u_2 v = uv_2 + 2u_1v_1 + u_2 v$

$y_3 = uv_3 + u_1v_2 + 2(u_1v_2 + u_2v_1) + u_2v_1 + u_3 v$

$= uv_3 + 3u_1v_2 + 3u_2v_1 + u_3 v$

$y_3 = uv_3 + {}^3c_1 u_1 v_{3-1} + {}^3c_2 u_2 v_{3-2} + u_3 v \quad [{}^3c_1 = 3,\ {}^3c_2 = 3]$

Since the theorem is true for $n=2$ & $n=3$ then it is true for any n Hence we can say

$(uv)_n = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2 + {}^nc_r u_{n-r} v_r$

or, $y_n = u v_n + {}^nc_1 u_1 v_{n-1} + \ldots + \cdots + u_n v$.

Successive differentiation

If y is a function of x, then its differential coefficient

$\frac{dy}{dx}$ is called first order derivative / differential coefficient of y.

In short, If $y = f(x)$, the successive derivatives are also denoted by

# Notation	First time	2nd time	Third time	n times
	$\frac{dy}{dx}$	$\frac{d^2y}{dx^2}$	$\frac{d^3y}{dx^3}$	$\frac{d^ny}{dx^n}$
	y_1	y_2	y_3	y_n
	y'	y''	y'''	$y^{(n)}$
	$\frac{d}{dx} = Dy$	D^2y	D^3y	D^ny
	$f'(x)$	$f''(x)$	$f'''(x)$	$f^n(x)$

The process of Differentiating the function again and again is called successive differentiation.

Obtain n'th derivaties of followings:

(1) $y = \frac{1}{x}$

(2) $y = \ln x$

(3) $y = x^n$

(4) $y = \cos x$

(5) $y = \sin x$

(6) $y = a^x$

(7) $y = e^{ax}$ (18 – DU)

(8) $\sin(ax+b)$ and $\cos(ax+b)$ – 18

(9) $\ln(ax+b)$

(3) Given that $y = x^n$, to show $D^n y = \lfloor n = n!$

$$y = x^n$$

Diff. w.r. to x

$$D^1 y = n x^{n-1}$$

$$D^2 y = n(n-1) x^{n-2}$$

$$D^3 y = n(n-1)(n-2) x^{n-3}$$

$$D^4 y = n(n-1)(n-2)(n-3) x^{n-4}$$

$$- - - - -$$

$$D^n y = n(n-1)(n-2)(n-3)(n-4) \cdots 4 \cdot 3 \cdot 2 \cdot 1 \cdot x^{n-n}$$

$$= \lfloor n \cdot x^0$$

$$= n! \cdot 1$$

$$= n!$$

$$\therefore D^n y = \lfloor n = n!$$

(2) $y = \ln x$

Diff w.r. to x

$$y_1 = \frac{1}{x} = x^{-1}$$

$$y_2 = (-) x^{-2}$$

$$y_3 = 2x^{-3} = (-1)^2 \lfloor 2 \, x^{-3}$$

$$y_4 = (-1)^2 (-3) \lfloor 2 \, x^{-4} = (-1)^3 \lfloor 3 \, x^{-4}$$

$$y_n = (-1)^{n-1} \lfloor n-1 \, x^{-n}$$

(iii) If $y = \frac{1}{x}$ Find y_n

Soln: Given that $y = \frac{1}{x}$

or, $y = x^{-1}$

Diff. w.r.to x

$$y_1 = (-1)\, x^{-2}$$

Again
$$y_2 = (-1)(-2)\, x^{-3}$$
$$= (-1)^2 \,\underline{|2}\, x^{-3}$$
$$y_3 = (-1)^2 (-3)\,\underline{|2}\, x^{-4}$$
$$= (-1)^3 \,\underline{|3}\, x^{-4}$$

— — — — — —

$$y_n = (-1)^n \,\underline{|n}\, x^{-(n+1)}$$

$$y_{n+1} = (-1)^{n+1} \,\underline{|n+1}\, x^{-(n+2)}$$

(iv) $y = \cos x$

$$y_1 = -\sin x$$
$$= \cos\left(\frac{\pi}{2} + x\right) \quad \left[\cos(90^\circ + \theta) = -\sin\theta\right]$$
$$y_2 = -\sin\left(\frac{\pi}{2} + x\right)$$
$$= \cos\left\{\frac{\pi}{2} + \left(\frac{\pi}{2} + x\right)\right\}$$
$$y_2 = \cos\left(\frac{2\pi}{2} + x\right)$$

$= a \cos\left[\frac{\pi}{2} + (ax+b)\right]$

Again, $y_2 = a^2\{-\sin(\frac{\pi}{2} + ax+b)\} = a^2\cos(2\frac{\pi}{2} + \dots)$

$y_3 = a^3\left[-\sin\{2\frac{\pi}{2} + (ax+b)\}\right]$

$= a^3\cos\left[\frac{\pi}{2} + 2\cdot\frac{\pi}{2} + (ax+b)\right]$

$= a^3\cos\left[3\frac{\pi}{2} + (ax+b)\right]$

$\therefore y_n = a^n\cos\left[n\frac{\pi}{2} + (ax+b)\right]$

(6) If $y = \ln(ax+b)$ show that $y_n = \frac{a^n(-1)^{n-1}\lfloor n-1}{(ax+b)^n}$ when $n \in N$

diff w.r. to x

$y_1 = \frac{a}{ax+b} = a\cdot(ax+b)^{-1}$

$y_2 = a^2(-1)(ax+b)^{-2}$

$y_3 = a^2\cdot a(-1)\cdot(-2)\cdot 1(ax+b)^{-3}$

$= a^3(-1)^2\lfloor 2\,(ax+b)^{-3}$

$\boxed{y_4 = a^4(-1)^2(-3)(ax+b)^{-3}}$

$y_4 = a^4(-1)^2(-3)\lfloor 2\,(ax+b)^{-4}$

$= a^4(-1)^3\lfloor 3\,(ax+b)^{-4}$

$y_n = a^n(-1)^{n-1}\lfloor n-1\,(ax+b)^{-n}$

$= \frac{a^n(-1)^{n-1}\lfloor n-1}{(ax+b)^n}$

More problems from H.S.C book.

Leibnitz Theorem.

State and prove Leibnitz theorem.

Gottfried Wilhelm Leibnitz (1646-1716)

Statement: If u and v are functions of x then the n^{th} derivative of the product of u and v is

$$(uv)_n = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2 + \cdots + {}^nc_r u_{n-r} v_r + \cdots + u v_n$$

where the suffixes of u and v indicate the order of differentiation w.r.to x.

Proof:- Let $y = uv$

$$\therefore y_1 = uv_1 + vu_1 = u_1 v + u v_1$$

Again differentiate

$$y_2 = u_1 v_1 + u v_2 + v_1 u_1 + v u_2$$

$$= u_2 v + 2u_1 v_1 + u v_2$$

Again differentiate,

$$y_3 = u_2 v_1 + v u_3 + 2(u_1 v_2 + v_1 u_2) + u v_3 + v_2 u_1$$

$$= u_3 v + 3u_2 v_1 + 2u_1 v_2 + u v_3$$

Since the theorem is true for $n=2$ and $n=3$ then it is true for any n. Hence we can say

$$(uv)_n = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2 + \cdots + {}^nc_r u_{n-r} v_r + \cdots + u v_n.$$

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Problem 1:

If $y = \tan^{-1}x$ show that $(1+x^2)y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0$

Sol^n:- Given, $y = \tan^{-1}x$

$$\Rightarrow y_1 = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2)y_1 = 1$$

Differentiating both sides n time by Leibniz theorem we have.

$$(1+x^2)y_{n+1} + {}^nC_1 y_n (2x) + {}^nC_2 y_{n-1} 2 = 0$$

$$\Rightarrow (1+x^2)y_{n+1} + 2nxy_n + \frac{n(n-1)}{2} y_{n-1}(2) = 0$$

$$\Rightarrow (1+x^2)y_{n+1} + 2nxy_n + n^2 y_{n-1} - ny_{n-1} = 0$$

(Hence proved)

Problem-2:

If $y = e^{a\cos^{-1}x}$ then show that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$$

Sol^n:- Given, $y = e^{a\cos^{-1}x}$

$$\Rightarrow \log y = \log e^{a\cos^{-1}x} \quad \Big| \quad \log e^x \Rightarrow x\log e$$

$$\Rightarrow \log y = a\cos^{-1}x$$

Diff. w.r. to x we get.

$\Rightarrow \left(\sqrt{1-x^2}\right) y_1 = -a y$

$\Rightarrow (1-x^2) y_1^2 = a^2 y^2$

$\Rightarrow (1-x^2) 2y_1 y_2 + y_1^2(-2x) = a^2 2y y_1$

$\Rightarrow (1-x^2) y_2 - x y_1 - a^2 y = 0$ [dividing both sides by $2y_1$]

$\Rightarrow (1-x^2) y_2 - x y_1 - a^2 y = 0 \quad \text{--- (1)}$

Differentiating both sides n times by Leibniz theorem we get.

$(1-x^2) y_{n+2} + n y_{n+1}(-2x) + \frac{n(n-1)}{\cancel{2}} y_n (-\cancel{2}) - x y_{n+1}$

$- n y_n - a^2 y_n = 0.$

$\Rightarrow (1-x^2) y_{n+2} - 2nx y_{n+1} - n^2 y_n + \cancel{n y_n} - x y_{n+1}$

$- \cancel{n y_n} - a^2 y_n = 0$

$\Rightarrow (1-x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2+a^2) y_n = 0$

(Hence proved)

H.W:- 3) If $y = e^{a\sin^{-1}x}$ prove that

$(1-x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2+a^2) y_n = 0$

Example

If $y = a\sin^{-1}x + b\cos^{-1}x$ then show that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$$

Sol^n:- Given that,

$$y = a\sin^{-1}x + b\cos^{-1}x \quad \text{--- (1)}$$

Diff (1) w.r.to x

$$y_1 = \frac{a}{\sqrt{1-x^2}} - \frac{b}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2}\, y_1 = a-b$$

$$\Rightarrow (1-x^2)y_1^2 = (a-b)^2$$

$$\Rightarrow (1-x^2)2y_1y_2 + (0-2x)y_1^2 = 0$$

$$\Rightarrow (1-x^2)y_2 - xy_1 = 0 \quad [\text{Dividing both sides by } 2y_1]$$

$\Rightarrow$ by using Leibniz theorem,

$$(1-x^2)y_{n+2} + {}^nC_1(-2x)y_{n+1} + {}^nC_2(-2)y_n - \{x y_{n+1} + {}^nC_1 \cdot 1\, y_n\} = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - 2nxy_{n+1} - \frac{n(n-1)}{2!}2y_n - xy_{n+1} - ny_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2-n+n)y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$$

[illegible]:

If $x = \sin\left(\frac{1}{m}\ln y\right)$ then

$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$.

Sol^n:- Given,

$$x = \sin\left(\frac{1}{m}\ln y\right)$$

$$\Rightarrow \sin^{-1}x = \frac{1}{m}\ln y$$

$$\Rightarrow m\sin^{-1}x = \ln y$$

$$\Rightarrow y = e^{m\sin^{-1}x} \quad \text{—(1)}$$

Diffue (1) w.r. to x

$$y_1 = e^{m\sin^{-1}x}\frac{m}{\sqrt{1-x^2}} \quad \text{—(2)}$$

$$\Rightarrow \sqrt{1-x^2}\,y_1 = my$$

$$\Rightarrow (1-x^2)y_1^2 = m^2y^2$$

$$\Rightarrow (1-x^2)2y_1y_2 - 2xy_1^2 = m^2 2yy_1$$

$$\Rightarrow (1-x^2)y_2 - xy_1 - m^2y = 0.$$

Leibniz theorem;

$$\Rightarrow (1-x^2)y_{n+2} + {}^nc_1(-2x)y_{n+1} + {}^nc_2(-2)y_n - \{xy_{n+1} + {}^nc_1(1)y_n\} - m^2y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - 2nxy_{n+1} - \frac{n(n-1)}{2!}2y_n - xy_{n+1} - ny_n - m^2y_n = 0$$

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$$\Rightarrow (1-x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2 - n + n + m^2)$$

$$\Rightarrow (1-x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2 + m^2) y_n = 0$$

* $x = 0$ [illegible] y_n [illegible] $(y_n)_0$.

Also find $(y_n)_0$.

$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 - n + n + m^2)y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$$

* If $y = f(x) = \sin^{-1}x$ then show that

(i) $(1-x^2)y_2 - xy_1 = 0$

(ii) $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2 y_n = 0$.

(iii) $(y_n)_0 = (n-2)^2 (n-4)^2 \ldots 3^2 . 1^2$ for n is odd numbers.

(iv) $(y_n)_0 = 0$ for n is even number.

Soln: Given, $y = \sin^{-1}x$ --- (i)

$$\Rightarrow y_1 = \frac{1}{\sqrt{1-x^2}} \quad \text{--- (ii)}$$

$$\Rightarrow y_1(\sqrt{1-x^2}) = 1$$

$$\Rightarrow y_1^2(1-x^2) = 1$$

$$\Rightarrow (1-x^2)2y_1y_2 - 2xy_1^2 = 0$$

$$\Rightarrow (1-x^2)y_2 - xy_1 = 0 \quad \text{--- (iii)}$$

Applying Leibnitz's theorem.

$$(1-x^2)y_{n+2} + n y_{n+1}(-2x) + \frac{n(n-1)}{2!} y_n(-2) - xy_{n+1} - ny_n \cdot 1 = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - 2xy_{n+1}n - n(n-1)y_n - xy_{n+1} - ny_n = 0$$

Mathematics

Mean value Theorem and Expansions

Chapter - 8 (4)

1. State and prove Rolle's theorem. (1691)

Statement:– If (i) $f(x)$ is continuous in the close interval $a \le x \le b$

(ii) $f'(x)$ ~~exists~~ differentiable i.e $f'(x)$ exists in the open interval $a < x < b$.

and (iii) $f(a) = f(b)$

then there exists at least one value of x (say c) between a and b [i.e, $a < c < b$], such that $f'(c) = 0$

Proof:– As the function $f(x)$ is continuous in the interval $a \le x \le b$, and $f(a) = f(b)$; $f(x)$ has at least a maximum or a minimum or both in the interval. Either $f(x)$ has maximum value M at $x = c$ i.e, $f(c) = M$ or a minimum value m at $x = c$ i.e, $f(c) = m$, (lying in the interval (a, b).

There are two cases:– (i) $M = m$

When $M = m$ then $f(x)$ be constant in the interval $[a, b]$ i.e, $M = m = f(x) = f(a) = f(b)$ and hence $f'(c) = 0$ when $a < c < b$

Mathematics
4th

Intregal theorem and Expansions

Chapter-3 (4)

1. State and prove Rolle's theorem. (1691)

Statement:- If (i) $f(x)$ is continuous in the close interval $a \le x \le b$

(ii) $f'(x)$ differentiable i.e $f'(x)$ exists ~~exists~~ in the open interval $a < x < b$.

and (iii) $f(a) = f(b)$

then there exists at least one value of x (say c) between a and b [i.e, $a < c < b$], such that

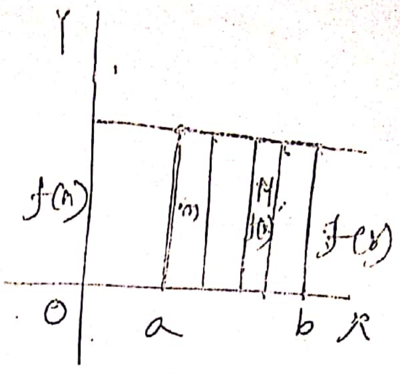
$f'(c) = 0$

Proof:- As the function $f(x)$ is continuous in the interval $a \le x \le b$, and $f(a) = f(b)$; $f(x)$ has at least a maximum or a minimum or both in the interval. Either $f(x)$ has maximum value M at $x = c$ i.e, $f(c) = M$ or a minimum value m at $x = c$ i.e, $f(c) = m$, c lying in the interval (a, b).

There are two cases:- (i) $M = m$

When $M = m$ then $f(x)$ be constant in the interval $[a, b]$ i.e, $M = m = f(x) = f(a) = f(b)$

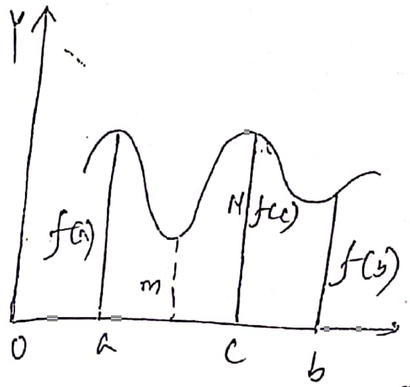
and Hence $f'(c) = 0$ when $a < c < b$



First case

<u>2nd case</u>:

When $M \neq m$, then at least one value of $f(x)$ is different from $f(b)$ and $f(a)$ in the interval. Let $f(c) = M$ be different from them, where c, lies within the interval.



Since $f(c)$ is the maximum value of the function $f(x)$, then $f(c+h) - f(c) \leq 0$, whether h is + or −.

Thus we have,

$\dfrac{f(c+h) - f(c)}{h} \leq 0$ when $h > 0$ and $\dfrac{f(c+h) - f(c)}{h} \geq 0$, when $h < 0$

But from the statement of the theorem we see that $f'(x)$ exists at $x = c$.

Now, $\lim\limits_{h \to 0} \dfrac{f(c+h) - f(c)}{h} \leq 0$ and

$\lim\limits_{h \to 0^-} \dfrac{f(c+h) - f(c)}{h} \geq 0$.

$\Rightarrow f'(c) \geq 0$ —— (1)

& $f'(c) \leq 0$ —— (2)

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[illegible] established.

Problem-1: Verify the truth of Rolle's theorem for the function $f(x) = x^2 - 3x + 2$ in the interval $(1, 2)$

Soln:- When $x = 1$, then $f(1) = 1 - 3 + 2 = 0$.

When $x = 2$, then $f(2) = 4 - 6 + 2 = 0$

$\therefore f(1) = f(2)$

Now, $f'(x) = 2x - 3$

If $f'(x) = 0$ then $2x - 3 = 0$

$\Rightarrow x = \frac{3}{2} = 1.5$

Thus we see that $f(x)$ is continuous in $1 \le x \le 2$; $f'(x)$ exists in $1 < x < 2$ and $f(1) = f(2)$. There exists a point $x = 1.5$ within the interval $(1, 2)$ such that $1 < 1.5 < 2$ where $f'(x) = 0$ i.e, $f'(1.5) = 0$

Problem-2: $f(x) = (x-2)(x-3)(x-4)$ in $2 \le x \le 3$

Soln:- Given, $f(x) = (x-2)(x-3)(x-4)$; $2 \le x \le 3$

or, $f(x) = x^3 - 9x^2 + 26x - 24$

or, $f'(x) = 3x^2 - 18x + 26$

$f(2) =$ [illegible]

$f(3) =$ [illegible]

Hence [illegible] satisfies the conditions [illegible]

Rolle's theorem [illegible] so there exists at least one point 'c' in $(2, 3)$, such that $f'(c) = 0$.

Now, $3c^2 - 18c + 26 = 0$.

$$\Rightarrow c = \frac{18 \pm \sqrt{18^2 - 12 \cdot 26}}{6}$$

$$= \frac{18 \pm \sqrt{324 - 312}}{6} = \frac{18 \pm \sqrt{12}}{6}$$

$$= \frac{18 \pm \sqrt{4 \cdot 3}}{6} = 3 \pm \frac{\sqrt{3}}{3}$$

$$= 3 \pm \frac{1}{\sqrt{3}}$$

$$= 3.57, 2.43$$

Clearly, $2.43 \in (2, 3)$.

Hence Rolle's theorem is verified.

Problem-3: verify Rolle's theorem

$f(x) = 2x^3 + x^2 - 4x - 2$.

Soln:- Given, $f(x) = 2x^3 + x^2 - 4x - 2$

Since, $f(x)$ is a polynomial in x,

so $f(x)$ is continuous and differentiable.

$2x^3+x^2-4x-2=0$

$\Rightarrow 2x^3+x^2-4x-2=0$

$\Rightarrow x^2(2x+1)-2(2x+1)=0$

$\Rightarrow (2x+1)(x^2-2)=0$

$\therefore x=-\sqrt{2}, -\frac{1}{2}, \sqrt{2}$

i.e., $f(-\sqrt{2})=(-\frac{1}{2})=f(\sqrt{2})=0$

Hence $f(x)$ satisfies the condition of Rolle's theorem. So there exists two points c_1 & c_2 & $f'(c_1)=0$ & $f'(c_2)=0$

$f'(x)=6x^2+2x-4=0$

$\Rightarrow 6x^2+6x-4x-4=0$

$\Rightarrow 6x(x+1)-4(x+1)=0$

$\Rightarrow x=-1, x=\frac{4}{6}=\frac{2}{3}$

Clearly, $c_1=-1\in(-\sqrt{2},-\frac{1}{2})$ &

$c_2=2/3\in(-\frac{1}{2},\sqrt{2})$

Hence Rolle's theorem is verified.

Statement :- If ... Closed interval $a \le x \le b$, $f'(x)$ exists in the open interval $a < x < b$, there exists at least one point c, $a < c < b$ such that $f(b) - f(a) = (b-a) f'(c)$

Proof:- let $\phi(x) = f(x) - Ax$ —— (i)

let $\phi(x)$ be an auxiliary fn, where A is Constant, so chosen that

$$\phi(b) = \phi(a)$$

i.e, $f(b) - Ab = f(a) - Aa$

$\Rightarrow f(b) - f(a) = Ab - Aa$

$$= A(b-a)$$

$$\Rightarrow A = \frac{f(b) - f(a)}{b-a} \quad \text{—— (ii)}$$

putting the value of A in eqn (i)

$$\phi(x) = f(x) - \left\{ \frac{f(b) - f(a)}{b-a} \right\} x \quad \text{—— (iii)}$$

Differentiating eqn (iii) w.r.to x.

$$\phi'(x) = f'(x) - \frac{f(b) - f(a)}{b-a} \quad \text{—— (iv)}$$

With the help of eqn (iii) and the given Conditions we find that $\phi(x)$ is continuous in $a \le x \le b$ and differentiable in

[illegible]

[illegible] the conditions of Rolle's theorem [illegible].

$\therefore$ Hence $\phi'(c) = 0$ where $a<c<b$. ---(iv)

From (iv) $\phi'(x) = f'(x) - A$

$\Rightarrow \phi'(c) = f'(c) - A$

$\therefore f'(c) - A = 0$

$\Rightarrow f'(c) = A$

$= \dfrac{f(b)-f(a)}{b-a}$

$\therefore f(b) - f(a) = f'(c)\,(b-a)$

Which is called first mean value theorem of Differential Calculus and Also known as Lagrange's form of the mean value theorem.

<u>problem:-</u> verify mean value theorem for the $f(x) = 3+2x-x^2$ in $(0,1)$.

<u>Soln:-</u> The given function, $f(x) = 3+2x-x^2$

Since $f(x)$ is polynomial, so $f(x)$ is continuous and differentiable in everywhere.

[illegible]

differentiable in [illegible]

∴ the mean [illegible]

here $a = 0$, [illegible]

But $f(a) = f(0) = 3$.

and $f(b) = f(1) = 3 + 2 \cdot 1 - 1 = 0$

$f'(x) = 2 - 2x$

$\therefore f'(c) = 2 - 2c$

So, we find by mean value theorem

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\Rightarrow 2 - 2c = \frac{4 - 3}{1 - 0} = \frac{1}{1} = 1$$

$$\Rightarrow 2 - 1 = 2c$$

$$c = \frac{1}{2} \in (0, 1)$$

Hence, the mean value theorem is valid for given f^n.

[illegible] Rolle's [illegible] function

$f(x) = 3x^3 + 7x^2 - 11x - 15$ [illegible]

Soln:- Given $f(x) = 3x^3 + 7x^2 - 11x - 15$

Since, $f(x)$ is a polynomial in x. so $f(x)$ is Continuous and differentiable in the $(-\infty, \infty)$

Here $f(x) = 3x^3 + 7x^2 - 11x - 15 = 0$

$= 3x^3 + 3x^2 + 4x^2 + 4x - 15x - 15 = 0$

$\Rightarrow 3x^3 + 3x^2 + 4x^2 + 4x - 15x - 15 = 0$

$\Rightarrow 3x^2(x+1) + 4x(x+1) - 15(x+1) = 0$

$\Rightarrow (x+1)(3x^2 + 4x - 15) = 0$

$\Rightarrow (x+1)(3x^2 + 9x - 5x - 15) = 0$

$\Rightarrow (x+1)\{3x(x+3) - 5(x+3)\} = 0$

$\Rightarrow (x+1)(x+3)(3x-5) = 0$

$\Rightarrow x = -1,\ x = -3,\ x = \frac{3}{5}$.

$\Rightarrow x = -1, -3, \frac{3}{5}$.

i.e, $f(-1) = f(-3) = f(3/5) = 0$

Hence $f(x)$ satisfies the condition of Rolle's theorem. so there exists two points c_1 & c_2 & $f'(c_1) = 0$ & $f'(c_2) = 0$.

Maclaurin's theorem

([illegible] differentiable function)

Statement:- Let $f(x)$ be a function of x which can be expanded in powers of x then

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \cdots + \frac{x^n}{n!} f^n(0) + \cdots$$

Proof:- let $f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots + a_n x^n + \cdots$ — (1)

Now successive differentiation w.r. to x we get, in eqn (1)

$$f'(x) = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots$$

$$f''(x) = 2 \cdot 1 a_2 + 2 \cdot 3 a_3 x + 3 \cdot 4 a_4 x^2 + \cdots$$

$$f'''(x) = 1 \cdot 2 \cdot 3 a_3 + 2 \cdot 3 \cdot 4 a_4 x + \cdots$$

- - - - - - - - - - - - - -

Now, putting $x = 0$ we get.

$$f(0) = a_0, \quad f'(0) = a_1, \quad f''(0) = 2!\, a_2, \quad f'''(0) = 3! \ldots$$

etc

Similarly, $f^n(0) = n!\, a_n$

putting all these values in eqn (1) we get;

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \cdots + \frac{x^n}{n!} f^n(0) + \cdots$$

$f(x) = \tan x$ $\qquad\qquad$ $f(0) = 0$

$f'(x) = \sec^2 x$ $\qquad\qquad$ $\therefore f'(0) = 1$

$f''(x) = 2\sec x \cdot \sec x \tan x$

$= 2\sec^2 x \cdot \tan x$ $\qquad \therefore f''(0) = 0$

$f'''(x) = 2\sec^4 x + 4\sec^2 x \tan^2 x$ $\qquad \therefore f'''(0) = 2$

By using Maclaurin's formula we get,

$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \cdots$

$\therefore \tan x = 0 + x \times 1 + \frac{x^2}{2!} \times 0 + \frac{x^3}{3!} \times 2 + \cdots$

$\therefore \tan x = x + \frac{2x^3}{3!} + \cdots$

$= x + \frac{x^3}{3} + \cdots$

(3) $\tan^{-1} x$

for $y = \tan^{-1} x$ $\qquad\qquad$ $\therefore f(0) = \tan^{-1}(0) = 0$

$f'(x) = \frac{1}{1+x^2}$ $\qquad\qquad$ $\therefore f'(0) = 1$

$= (1+x^2)^{-1} = 1 - x^2 + x^4 - x^6 + x^8 \cdots$

$f''(x) = -2x + 4x^3 - 6x^5 + \cdots$ $\qquad \therefore f''(0) = 0$

$f'''(x) = -2 + 12x^2 - 30x^4 + \cdots$ $\qquad \therefore f'''(0) = -2$

$f^{iv}(x) = 24x - 120x^3 + \cdots$ $\qquad \therefore f^{iv}(0) = 0$

e^{mx}

$f(x) = y = e^{mx}$; $f(0) = e^0 = 1$

$f'(x) = me^{mx}$ $f'(0) = me^0 = m$

$f''(x) = m^2e^{mx}$ $f''(0) = m^2e^0 = m^2$

$f'''(x) = m^3e^{mx}$ $f'''(0) = m^3e^0 = m^3$

$e^{mx} = 1 + mx + \frac{m^2x^2}{2!} + \frac{m^3x^3}{3!} + \ldots$

$e^x \sin x$

$f(x) = e^x \sin x$ ∴ $f(0) = 0$

$f'(x) = e^x(\cos x + \sin x)$ ∴ $f'(0) = 1$

$f''(x) = e^x(-\sin x + \cos x) + (\cos x + \sin x)e^x$

$= -\sin x\, e^x + e^x \cos x + \cos x\, e^x + e^x \sin x$

$= 2e^x \cos x$ ∴ $f''(0) = 2$

$f'''(x) = -2e^x \sin x + 2e^x \cos x$ ∴ $f'''(0) = 2$

$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \ldots$

Taylor's theorem:

Statement:- Taylor series of degree n about a is

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

This formula $f(x)$ near a.

* Find the third degree Taylor polynomial of $f(x) = \sqrt{x}$ with center $x_0 = 0$.